

10th International Young Scientist Conference on Computational Science (YSC 2021)

Review and comparison of prediction algorithms for the estimated time of arrival using geospatial transportation data

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Abstract

Considering the increasing number of vehicles used for a variety of purposes, the demand for accurate traffic services is also rising. Most drivers use traffic services to find a path through the city and get an estimation of how long the trip will take. Due to a large number of factors affecting driving in urban areas the task of predicting the estimated time of arrival is not trivial. In this paper we discuss different methods used for trip duration prediction and compare their performances. We use a real urban traffic data set for testing, and also make this data set public to stimulate future research.

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Peer-review under responsibility of the scientific committee of the 10th International Young Scientists Conference on Computational Science

Keywords: routing; ETA prediction; urban traffic; routing machine

1. Introduction

Despite the ever-growing quality of modern public transportation systems, there are still many people who regularly commute using their own vehicles or carsharing services [6]. With the growing number of cars in urban areas the need of efficient and accurate traffic services becomes more and more crucial. One of the most frequently used services among drivers is *routing*, i. e. the task of finding a path between two or more points on the Earth's surface.

Each day drivers use routing services to look for the quickest path through the city. One of the key components in such services is providing a user with an *estimated time of arrival* (or *ETA*), which is an estimate on the duration of a planned trip.

In days of pandemic delivery services received a huge boost [9, 8] which led to expansion of the concept of ETA far beyond the scope of urban navigation. Delivery-oriented businesses are interested in providing their customers with an accurate ETA to increase the quality of service. Outside the area of food and goods delivery, many Business-to-Business companies are also interested in accurate ETA since latency in logistics may lead to substantial material

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losses. On the other hand, ETA is equally prominent from the scientific standpoint, as some researchers are also investigating the effects of traffic congestion on people's health [7, 16, 15]. Accurate ETA allows a driver to better understand what the trip will be, better organize their time, and it can even nudge drivers into changing their plans if the ETA is too large.

Considering the complexity of such a large and intricate system which is a modern city, accurate prediction of ETA becomes a rather challenging task. Indeed, many factors greatly affect the traffic situation in the city, such as weather [5], configuration of the city streets [33], actions and behavior patterns of the driver and the other participants of the traffic [2] and so on. The relations between these variables are not always clear. Despite the fact that traffic flow simulations are widely used in different areas of urban science, the majority of them do not have capability to accurately simulate traffic in real time [13]. Some of the existing simulators cannot perform real-time simulation, some are difficult to install and setup, others do not allow to perform routing in the simulations. As an result, machine learning algorithms have become more and more popular for ETA prediction.

Even though there are many available methods for ETA prediction (see Section 3), there have not been any attempts to compare them all on a common data set. The main reason for this is the lack of such data sets in public domain: in most cases, researchers use some privately available data in order to compare their method to some baseline (which is also not unified in literature), therefore comparing different methods based on the authors' findings becomes an onerous task.

The main aim of this work is to fill this gap by comparing several existing machine learning methods of ETA prediction on a common data set. The data is the real-world traffic flow data collected by StarLine Ltd. in St. Petersburg, Russia. We also make a part of this data set available for the general public in order to stimulate future research and experiments in this field¹.

The rest of the paper is structured as follows. Section 2 is dedicated to discussion on existing papers which address the ETA prediction task. In Section 3 we briefly describe some of the core methods used for ETA prediction. The experimental results as well as information on the used data and method implementations is presented in Section 4. Finally, we conclude the paper and state our thoughts on future research in Section 5.

2. Related work

In this paper we are mainly interested in methods of ETA prediction for personal transportation, such as private vehicles or carsharing services. To the best of our knowledge, there has not been any prior attempts to survey and compare different methods of ETA prediction in this case. Nonetheless, we mention an existing paper [21], in which the authors compared different ETA prediction methods for buses. This paper focuses on variety of methods for ETA prediction of public transport, especially on Machine Learning methods. Authors do not conduct their own experiments and emphasize the fact that objective comparison of different methods is difficult due to lack of standards of evaluation and data sets. A number of reviewed approaches exploit data which is available only in case of busses' ETA prediction such as information from other vehicles from the route, timetables and locations of bus stops.

The majority of existing methods for ETA prediction use GPS data collected from vehicles. Each data record consists of GPS coordinates of a vehicle along with a timestamp of the measurement and the unique identifier of the device. This information allows to group GPS points into ordered tracks and fully describe the trajectory of a vehicle. Considering wide variety of factors which affect ETA, additional information such as weather, current traffic and historical traffic can also be used for ETA prediction.

The availability of different possible data sources allows to adapt various non-ETA methods to the problem of ETA prediction. Some of these use historical data in order to predict the state of the traffic for a given time [11], others rely on statistical approaches [23, 22]. Methods such as Kalman filter may yield appropriate results [21, 18], however most of them require accurate information about traffic on a route which in most cases is unavailable in case of general purpose routing.

Due to the growing popularity of big data and machine learning approaches in various fields of science and technology, many researchers attempt to apply machine learning techniques to solve traffic-related problems. Some of

¹ Data set is available via link: <https://github.com/RamiNaim/starline-traffic-data>

machine learning solutions describe a route as a feature vector and then treat the problem of ETA prediction as a regression problem. Nowadays, Random forests, Support vector machines, Gradient boosting, Neural networks and other methods are successfully applied for predicting ETA [26, 20, 28, 30]. Moreover, considering the availability of spatiotemporal information in form of GPS points and timestamps, more advanced methods of handling such data are being developed. Several solutions [27, 24] based on artificial networks use Long short-term memory networks and special Geospatial convolutional layers for effective raw GPS data processing. However, most of them require large amounts of data and are computationally expensive.

3. Methods of ETA prediction

The overall task of ETA prediction in this paper is as follows: given the known variables describing the traffic state at the moment and the route, predict the duration of the trip. The path searching stage is not considered here and provided by a routing engine.

Below are some of the existing approaches to solving this task.

3.1. Historical average speeds

Historical Average Speeds (HAS) is often used as a baseline method for ETA prediction task [21, 11, 14]. This approach implies calculating average speeds of all vehicles travelling along a certain road segment at a certain time. Average speeds are calculated over a certain period, e. g. during a year. The time intervals can be picked arbitrarily, for instance these may be 10 minute intervals of a day, a week, a month or a year. The use of historical data results in average characteristics of a traffic for a specific road and does not take into account any additional information. Consequently, it can yield poor results, for instance, in case of car accidents on the road which are expected in a heavily-loaded city.

3.2. Random forest regression

Random forest regression (RFR) is an ensemble method used for solving regression problems. RFR trains group of decision trees, or weak predictors, on different subsets of data and make a prediction based on average predictions of those predictors. It is also sometimes treated as a baseline method for prediction since it generally can cope with smaller data sets for training compared to other approaches [21, 10, 31]. Considering the lack of traffic data sets and costly geospatial computations, the ability to generate adequate prediction from limited amount of data makes RFR a valid candidate for ETA prediction. This method was applied in [29].

3.3. Support vector regression

Support vector regression (SVR) is a regression algorithm that is based on Support vector machines used for classification. The training phase of SVR consists of building a hyperplane in the feature space which minimizes the loss function. Similar to RFR, SVR is able to provide reliable predictions using relatively small training data sets. This method was applied in [28, 17].

3.4. Gradient boost regression

Gradient boost regression (GBR) is a boosting algorithm that uses a collection of weak predictors (typically, decision trees) to make a prediction. The weak predictors are fitted sequentially based on the gradient of chosen loss function to minimize the overall quality of the final prediction. Different types of GBR algorithms are used for ETA prediction in several works and show relatively good results. This method was applied in [26].

3.5. Neural networks

Neural networks (NN) are also widely used and quite popular for ETA prediction, since these methods are able to process data with large numbers of features (which is usually the case when constructing features of routes). This method was applied in [1, 25].

3.6. Geospatial networks

As an evolution of the Neural network approach, Deep neural networks with LSTM and special Geo-convolutional layers are also employed [12]. A notable example of such approach is the DeepTTE method [27]. Approaches like this allow inputs of arbitrary length and process raw GPS data. Geo-spatial networks often require much larger data sets compared to other solutions.

4. Experiments

4.1. Compared methods and metrics

Some of the methods from Section 3 were chosen for the experimental research, namely, RFR, GBR, SVM (using the implementations from Python's scikit-learn package [19]), NN (using the implementation from Python's keras package [3]), and finally, DeepTTE (using their authors' implementation [4] available on github).

The parameters of the compared methods are present in Table 1.

Table 1. Parameters of compared methods.

Model	Parameters
RFR	# of estimators = 20, max depth = 5
GBR	# of estimators = 200, max depth = 10
SVR	6-degree polynomial kernel, C = 1
NN	2 hidden layers with 8 neurons with relu activation, loss function = MAE, 40 epochs
DeepTTE	kernel size = 3, $\alpha = 3$

The aforementioned approaches are evaluated using standard metrics, such as Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) and standard deviation.

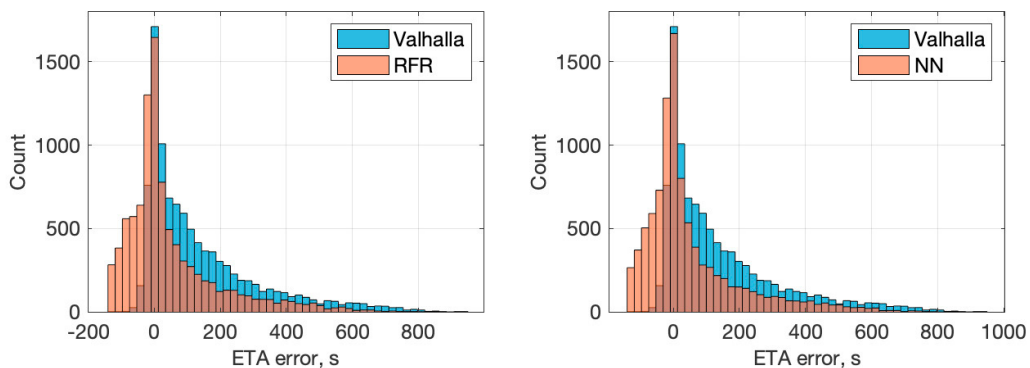


Fig. 1. Histogram of RFR (left) and NN (right) errors, compared with the errors of the Valhalla approach. The errors are calculated in seconds.

4.2. Data set

As it was mentioned previously, the lack of standard data sets is one of the big issues regarding the ETA prediction problem. In this paper we use the transportation data provided by StarLine Ltd., which we also make public.

The data received from StarLine Ltd. comes in a form of messages from GPS devices located on vehicles. Each message includes the location of the vehicle (latitude and longitude in WGS-84 projection), speed, timestamp, and unique identifier of the vehicle. In total there are 298089 tracks collected during one weekday grouped by vehicle ID. The tracks are up to 2 kilometers long. The real duration of a trip is acquired from timestamps of first and last GPS points comprising the track. This duration is used for calculating MAE, MAPE and standard deviation.

Raw GPS data generally requires some form of preprocessing, most frequently map-matching, i.e. the process of finding the best sequence of road segments which represents the sequence of raw GPS points. For this purpose a *routing machine* can be used. Routing machines (or routing engines) are special frameworks which perform shortest path search on road graphs, i.e. graphs that represent the map of the area's roads and streets. One of the most used map data source used is the OpenStreetMap project [32]. The OSM map data is fed into the routing engine in order to find a route between two or more points or match raw GPS data to a routing graph. In this work, the Valhalla² routing engine is used.

The DeepTTE solution requires raw GPS data aggregated in tracks, i.e. sequences of latitudes, longitudes, time of the trip, distance and some additional information. It also requires the distance of a route. Arrays of latitudes and longitudes of each track are then processed by Valhalla routing service in order to receive the distance of the whole route along the road graph.

By contrast, the GBR, NN, SVR and RFR methods are designed for solving regression problems, and consequently, they require input data in form of feature vectors. Due to the imprecision of GPS devices, the received points generally do not align with the routing graph. Therefore we first apply the map-matching step, as described above. The data is aggregated in tracks using the unique vehicle IDs and then fed into Valhalla routing engine which matches the tracks to a road graph. This expands the data set, adding to raw points the road segments on which the vehicle was when the measurement was done.

Those GPS points may be grouped by the road segment they are assigned to. This allows to calculate average vehicle speed on a segment, based on vehicles which pass it during the last 10 minutes and estimate the traffic on the available roads of the city. In many works related to traffic and ETA prediction the authors frequently use the term congestion index r . The congestion index is calculated as a ratio between the current speed of vehicles on the road segment and the free speed v_{free} of the segment (when there is no congestion and vehicles are freely passing through it). Historical data is used to estimate v_{free} as average speed of vehicles on the road segment late at night. Based on the value of the congestion index, each road segment is classified as:

- jammed, if $r \in [0, 0.25)$,
- slow, if $r \in [0.25, 0.5)$,
- normal, if $r \in [0.5, 0.75)$,
- free, if $r \in [0.75, \infty)$.

When the classification is done, for each 10-minute interval of a day there is a list of road segments, each having a class assigned to it. Knowing the road segments constituting the path, we define how congested the path is by counting the road segments of each class. These values are used as features which describe the trips in the data set.

The distance of each track is then acquired in the same way as it was described previously for the DeepTTE data set. Additionally, the Valhalla routing service is used to calculate the ETA of each track, based purely on internal constant weights of the Valhalla's routing algorithm. Such ETAs are thus generally not very accurate, but can also provide valuable information for more accurate ETA prediction.

The resulting feature vector describing each trip includes distance of a trip, Valhalla's ETA, hour when the trip is started and the number of road segments of each congestion class.

² Valhalla documentation and code is available via link: <https://github.com/valhalla/valhalla>

In contrast to a number of other methods of GPS data preprocessing, mentioned in works from [21], we do not make any assumptions about the path, its length or vehicles previously traveled along this route. To our knowledge, there was no attempt to use classified congestion indexes to describe traffic information of an arbitrary path.

For the experiments the data was split into training and testing samples in 80 to 20 proportion.

Since no historical data is available to implement HAS approach, the training data set was used to calculate the average speeds for roads in the graph for each 10 minutes of a day as it was described above. The ETA was then computed as sum of time spent to move along the road segments constituting the path with given average speeds.

4.3. Results

Based on the predictions acquired for the test data set a distribution of ETA prediction error was computed. The error is defined as difference between the real duration of a trip and the predicted ETA value. Additionally, for a reference the distribution of Valhalla's routing service errors was also computed.

The results for the RFR and NN methods can be seen on Fig. 1. Shown on Fig. 2 are the results for the SVR and GBR methods. Finally, the results for the DeepTTE and HAS methods are shown on Fig. 3.

It can be seen from the figures that the majority of ETAs computed by Valhalla is shorter than the real duration of trips, whereas for the rest of the considered methods the mean error is shifted closer to zero.

The numerical metrics for each method are present in Table 2. We can see that DeepTTE shows the worst performance compared to other solutions. It can be explained by the fact that the used data set is much smaller than the data set used by the method's authors. The SVR method seems to perform the best on the proposed data set, scoring lowest in both MAE and MAPE, although the results for the rest of the regression methods are not very different. The lowest standard deviation is achieved using the RFR method, which is expected from an ensemble method.

Table 2. Evaluation results

Method	MAE	MAPE	Standard deviation
Valhalla	143.3542	37.5144	172.4718
RFR	97.5038	33.0446	127.8062
GBR	94.1070	31.7819	145.6232
SVR	91.6505	29.9846	145.2561
NN	97.2522	31.6609	149.8864
DeepTTE	185.0625	102.3828	239.5208
HAS	166.6192	54.7373	215.8854

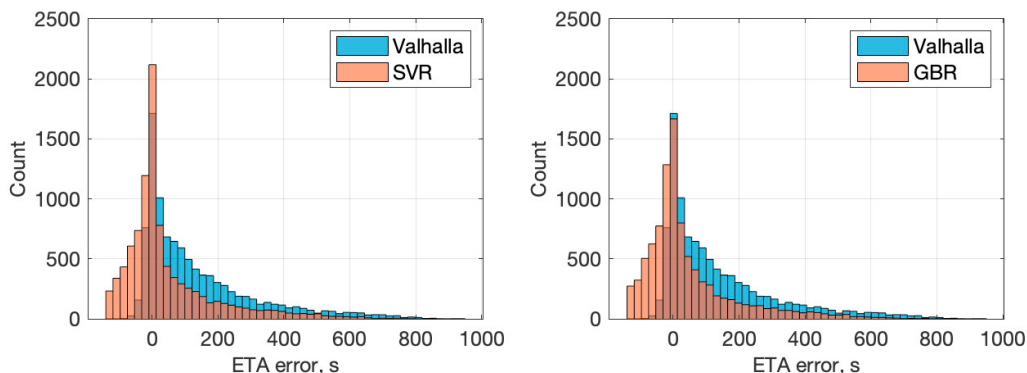


Fig. 2. Histogram of SVR (left) and GBR (right) errors, compared with the errors of the Valhalla approach. The errors are calculated in seconds.

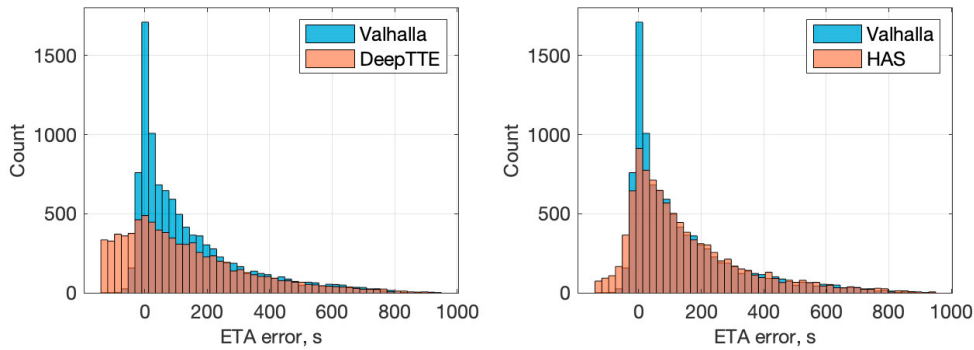


Fig. 3. Histogram of DeepTTE (left) and HAS (right) errors, compared with the errors of the Valhalla approach. The errors are calculated in seconds.

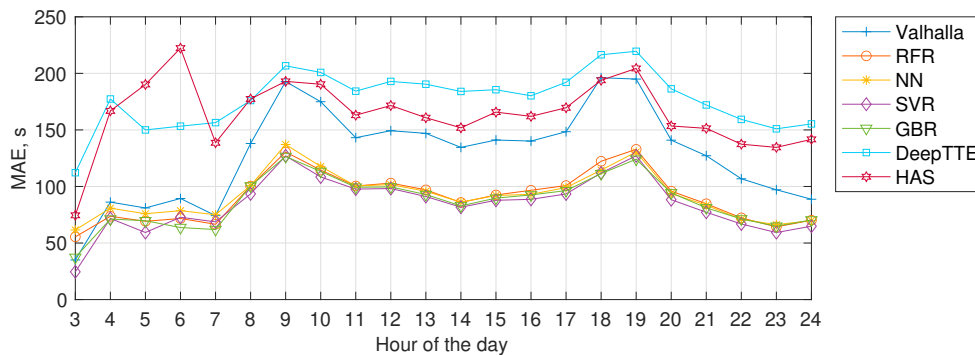


Fig. 4. Error distributions of various methods during different times of the day.

On Figure 4 a graph with mean errors for each algorithm during a day is present. As it can be seen, the prediction error changes during a day being maximum at the rush-hours. This is an expected outcome since during rush-hours there are typically the largest amount of vehicles in the city, and the traffic state could change unexpectedly.

5. Conclusion and future work

During this work we compared several core algorithms used for ETA prediction. Additionally, two baseline methods such as Valhalla routing service and historical average speeds were used to provide reference values for comparison. The numerical results of the experiments are present in Table 2.

All methods except of DeepTTE showed similar behavior and managed to reduce mean error as well as standard deviation. Only SVR managed to get MAPE to less than 30%, also providing the best MAE of 91.65 seconds. DeepTTE showed the worst results among all methods. This could be explained by the fact that authors of it suggest to use much bigger data sets to receive appropriate results.

Motivated by this need for larger data sets, an interesting idea for future research would be synthetic traffic data generation, based on real traffic data. This could both allow the use of more sophisticated solutions for the ETA prediction task, that need larger input data, and also address the need for open traffic data sets, as synthetic data is usually much easier to make public.

Acknowledgements

This research is financially supported by The Russian Science Foundation, Agreement №17-71-30029 with co-financing of Bank Saint Petersburg.

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